ELECTRONICS · CHART 2

ESP32-C3

HazardGuard C3

Wiring, the program's decisions and our test results

Reads every sensor 4 times a second · limits saved through power cuts

1 Wiring

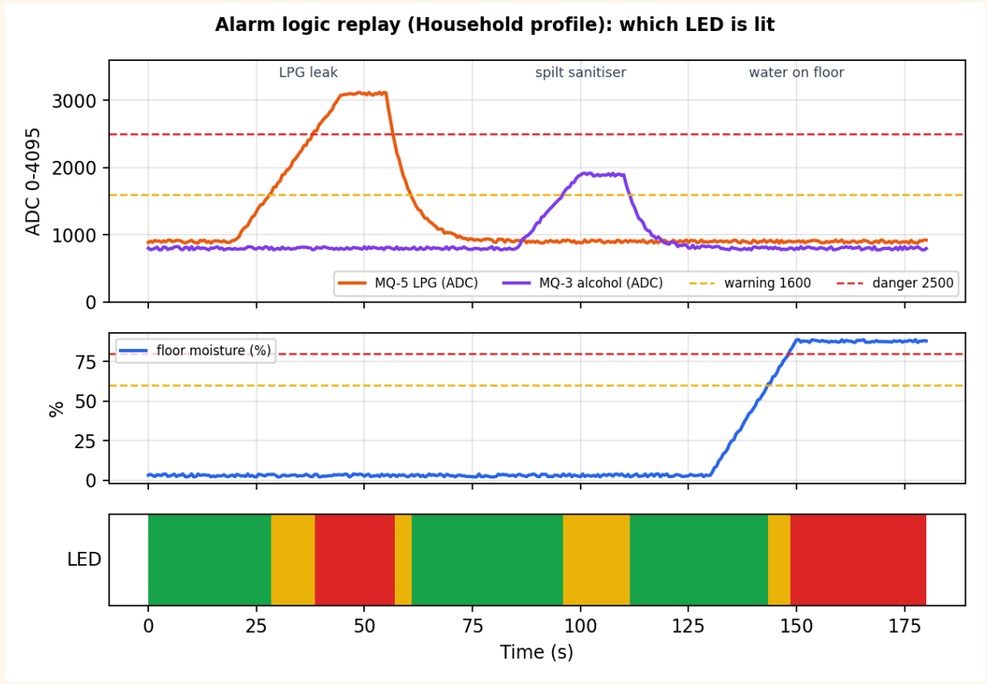
Part	ESP32-C3 pin	Power
MQ-5 AO	GPIO 1	5 V
MQ-3 AO	GPIO 4	5 V
Soil probe AO	GPIO 0	3.3 V
DHT11 DATA	GPIO 6	3.3 V
Green LED	GPIO 7	220 Ω to GND
Yellow LED	GPIO 8	220 Ω to GND
Red LED	GPIO 3	220 Ω to GND

Analog sensors use ADC1 (GPIO 0-4): ADC2 stops working while WiFi is on. GPIO 2 and 9 stay empty - they are boot pins.

2 What the program does

1. Read MQ-5, MQ-3, soil (average of 8)
2. Read DHT11 every 2 s
3. Pin floating? -> 'not connected'
4. Warming up? no gas alarm for 60 s
5. Compare with the profile's limits
6. Danger -> red strobe
7. Warning -> yellow blink
8. Safe -> green, update phone

3 Test: leak, spill and flood



The firmware's own logic replayed. The bottom strip shows which LED was lit.

4 Bench tests

We did	It showed
Unlit gas lighter at MQ-5	yellow, then red
Sanitiser near MQ-3	yellow, then red
Hot air on DHT11	red: OVERHEAT
Probe in water	red: WATER LEAK
Unplugged a sensor	'not connected'

5 Clever bits

- **Unplugged check:** pull-up, then pull-down - a loose pin swings wildly.
- **3 profiles:** Household, Hotel kitchen, Industrial.
- **LED test** at power-up and from the phone.

6 Phone dashboard

- Join WiFi **HazardGuard-C3** (no password).
- It opens **192.168.4.1** by itself.
- Live gauges, history, LED mirror, limits.

7 Next steps

Relay to start an exhaust fan or close the gas valve · buzzer and flame sensor · SMS alerts · battery backup